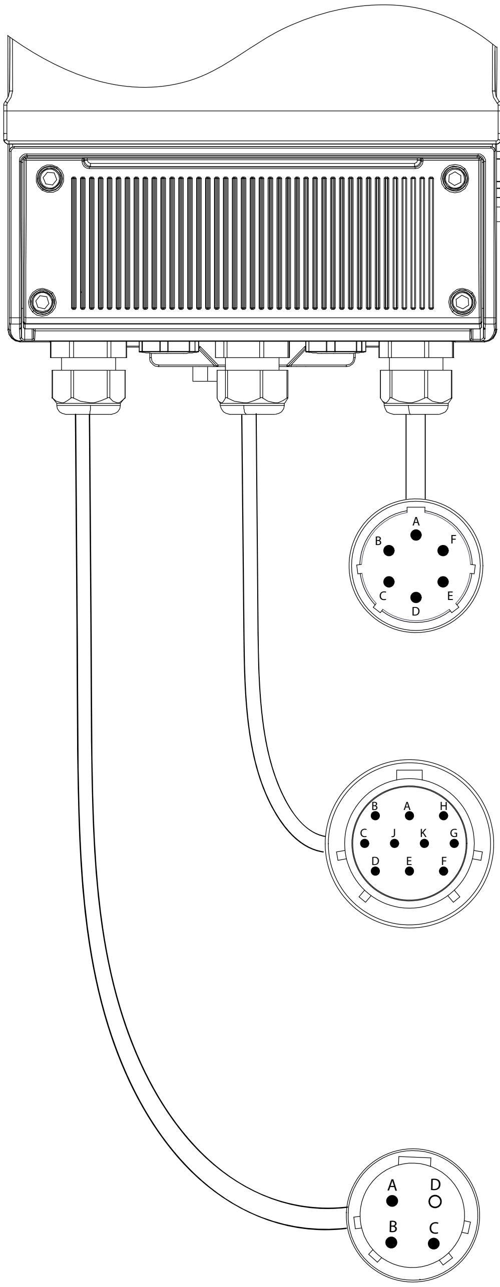
CONNECTORS MIL

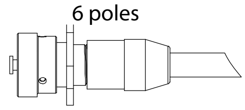
THE following are the links of the MIL connectors IP68



SENSOR SIGNALS

A contact from terminal 1 of the converter (electrode 1)
Contact F from terminal 2 of the converter (electrode 2)
Contact And from terminal 3 of the converter (COM. Elec.)
Contact B from terminal 13 of the converter (COIL 1)
Contact C from terminal 12 of the converter (COIL 2)
Contact D from terminal 4 and 11 of converter
(SHIELD electrodes - COILS)

Connector
6 poles

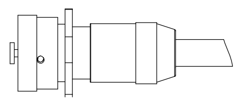


INPUT/OUTPUT

1. Contact B from terminal 24 of the converter (Out1 4-20 +)
2. Contact A from terminal 25 of the converter (Out2 4-20 +)
3. Contact H from terminal 23 of the converter (Out1-2 4-20 -)
4. Contact C from terminal 21 of the converter (+ 24V)
5. Contact J from terminal 22 of the converter (0V)
6. Contact K from terminal 15 of the inverter (IN1 +)
7. Contact G from terminal 16 of the converter (IN1-)
8. Contact D from terminal 19 of the converter (C Out2)
9. Contact E from terminal 18 of the converter (Out1 C)
10. Contact F from terminal 17 of the converter (Out1-2 E)
11. Contact C from terminal 28 of the converter (RS485 A)
12. Contact J from terminal 29 of the converter (RS485 B)

NB: the connections 4 and 5 exclude the use
connections 11 and 12.

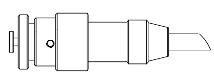
Connector
10 poles



POWER SUPPLY

C from terminal L power converter (+ dc)
A from terminal N power converter (-dc)
B from terminal GROUND Power Converter

Connector
4 poles

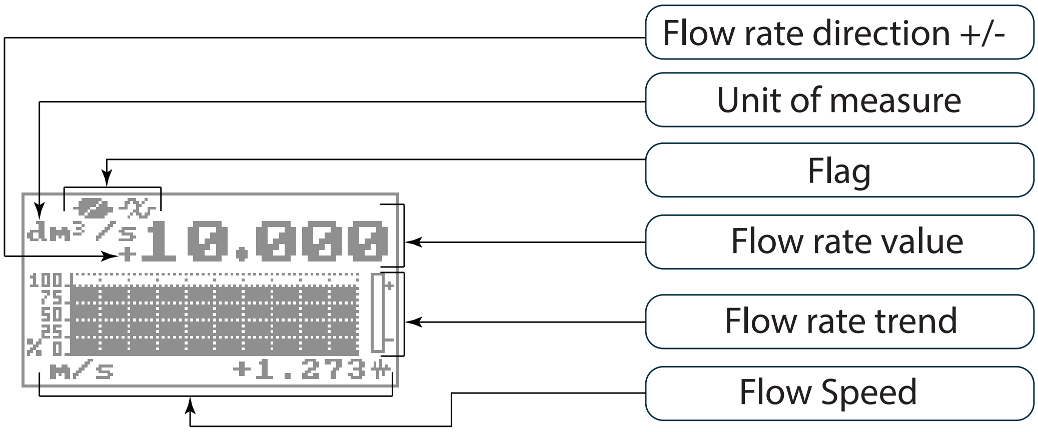


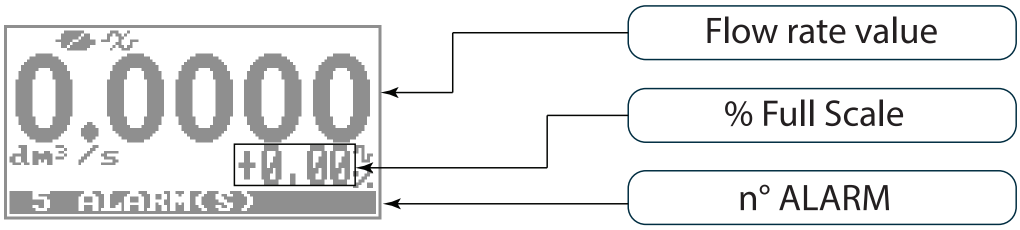
NOTE: Military Connector 6 poles for sensor converter only provided in the separate version of the converter.

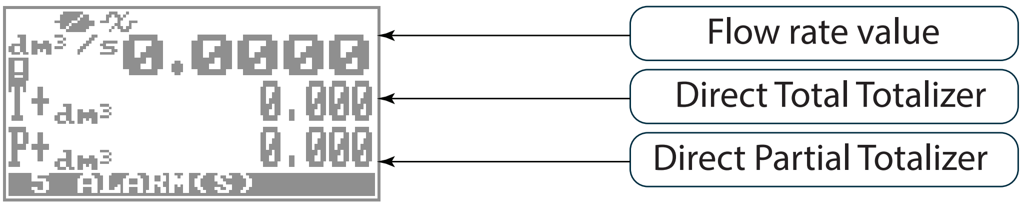
START VISUALIZATION PAGES

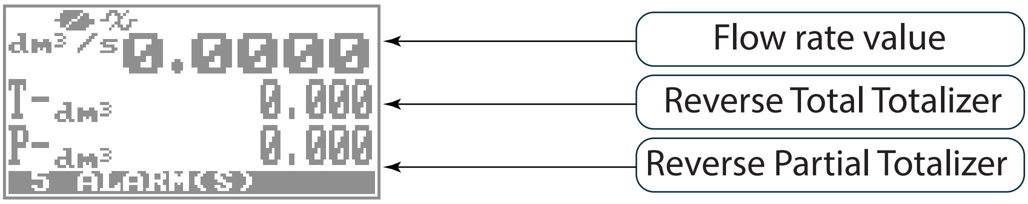
The direct exposure of the converter to the solar rays, could damage the liquid crystal display.
The visualization pages can be change according to instrument's setup.

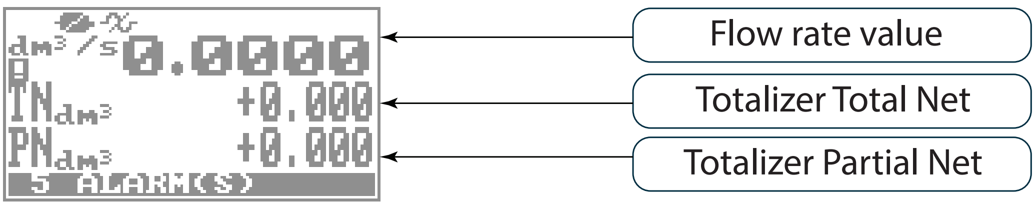
Push to change visualization

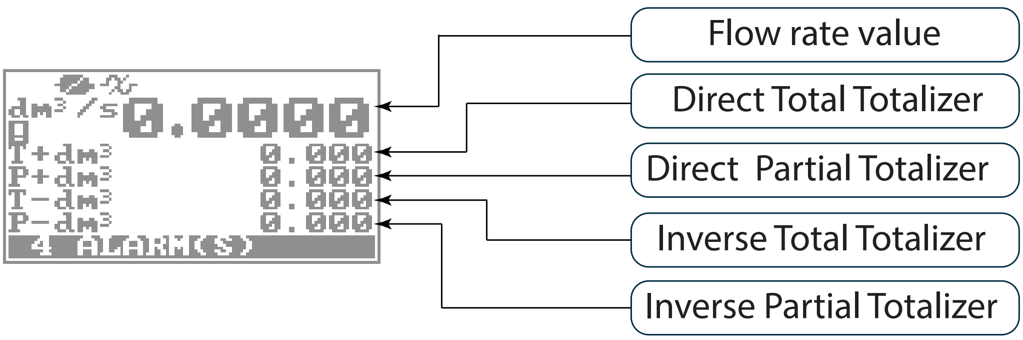


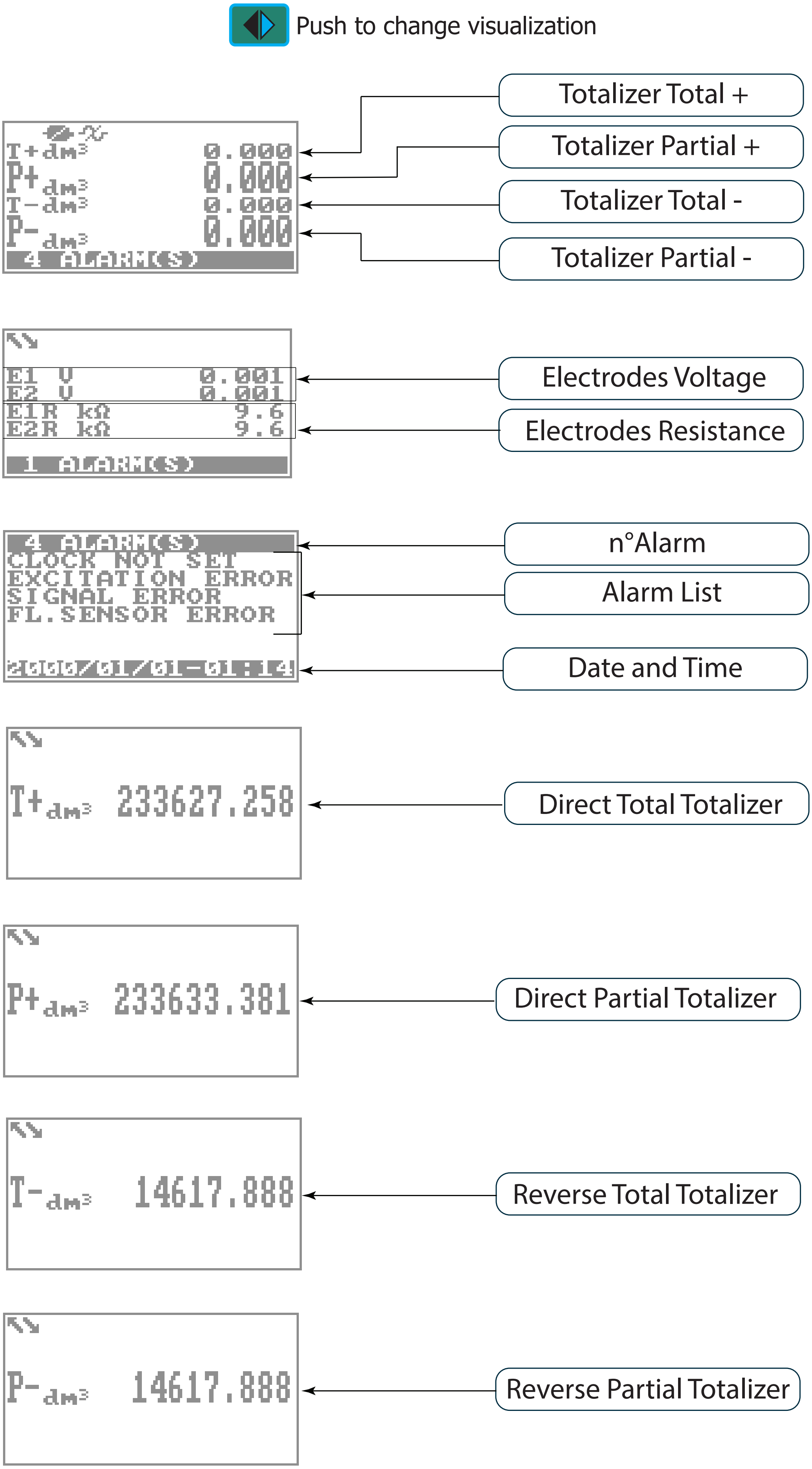










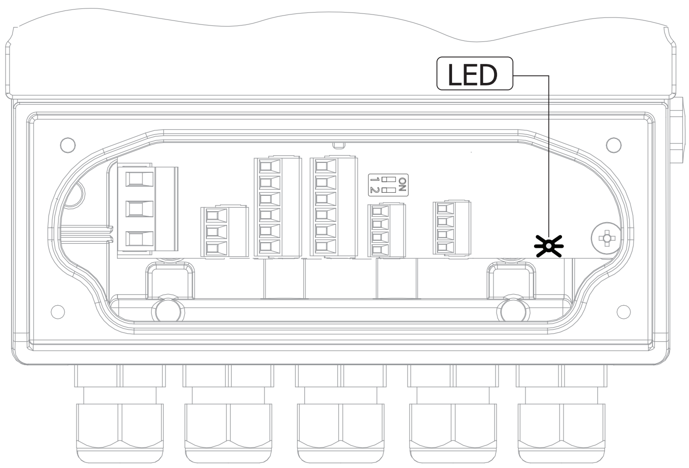


The visualitation page about batching is described in the dedicated section "BATCHING" pag45

MEANING OF FLAGS

FLAG	DESCRIPTION	FLAG	DESCRIPTION
	EMPTY PIPE		MIN FLOW ALARM
	FILE UPLOAD		MAX FLOW ALARM
	FILE DOWNLOAD		VIDEO TERMINAL CONNECTED
	BATTERY RECHARGE (FLASHING) LOW BATTERY (FIXED)		FLOW RATE OVERFLOW
	FLOW RATE SIMULATION (FLASHING)	1	PULSE 1 OVERFLOW
	CALIBRATION (FLASHING)	2	PULSE 2 OVERFLOW
	GENERIC ALARM (FLASHING)		POWERED DEVICE WITH ONE CHARGERS BATTERY (MID-DIRECTIVE) oppure BATCHING IN PROGRESS
	GENERAL ALARM ONLY ON PHYSICAL DISPLAY (FLASHING)		
	SIGNAL ERROR		
	EXCITATION ERROR		

MEANING OF LED COLORS

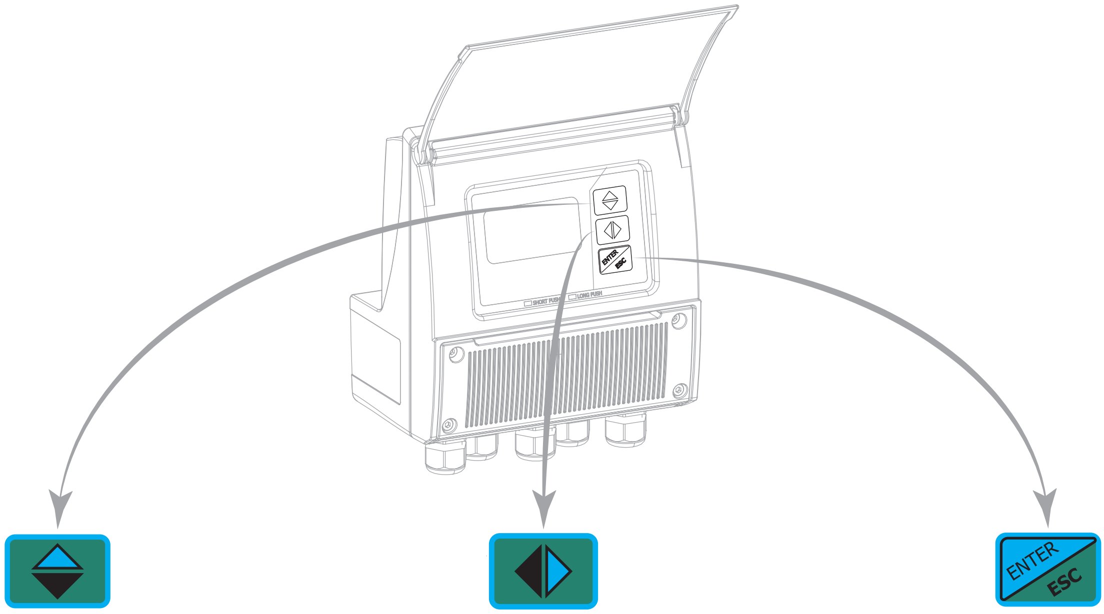


LED Red: Alarm signal	
LED Blue: Usb communication enable	
LED Green: Functioning system correctly	

ACCESS TO THE CONFIGURATION MENU

The configuration can be done in two different ways:
- ❑ By keypad of converter
- ❑ By MCP interface (Virtual display of instrument)

ACCESS VIA KEYPAD



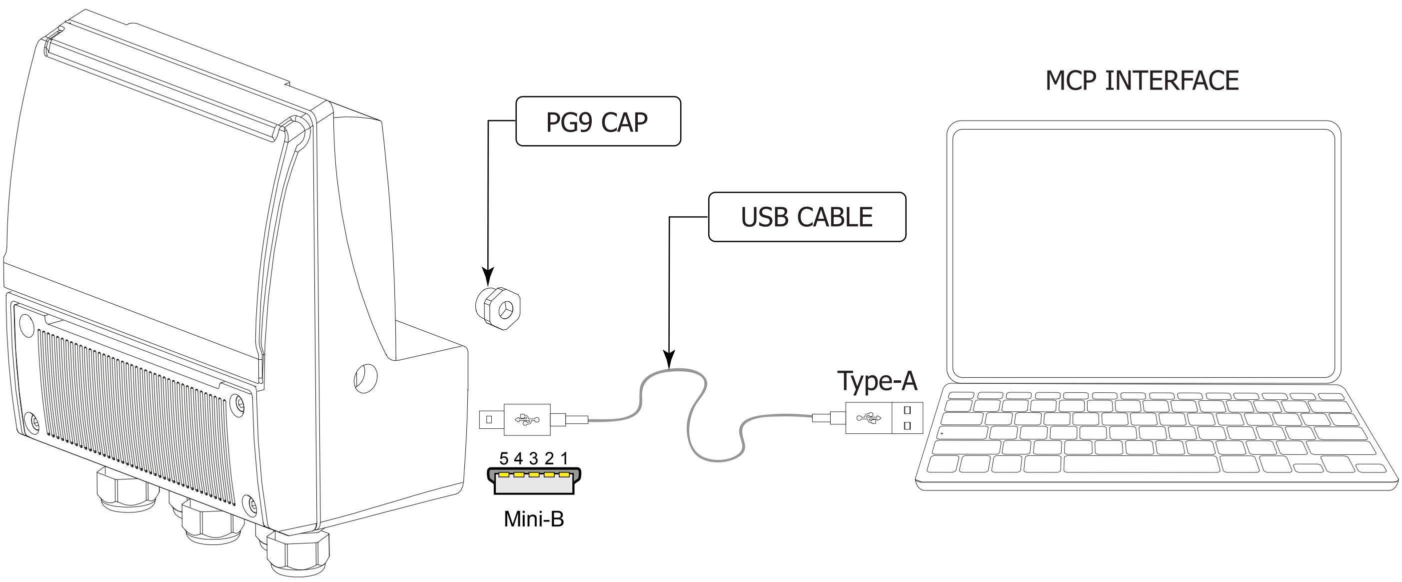
SHORT PRESSING (< 1 SECOND):
Increases the numeric figure or the parameter selected by the cursor
Returns to the previous subject on the menu.
LONG PRESSING (> 1 SECOND):
Decreases the numeric figure or the parameter selected by the cursor. Proceeds to the next subject on the menu.

SHORT PRESSING (< 1 SECOND):
Moves/positions the cursor rightward on the input field. Proceeds to the following subject of the menu. Change the display of the process data
LONG PRESSING (> 1 SECOND):
Moves/positions the cursor leftward on the input field. Returns to the previous subject on the menu

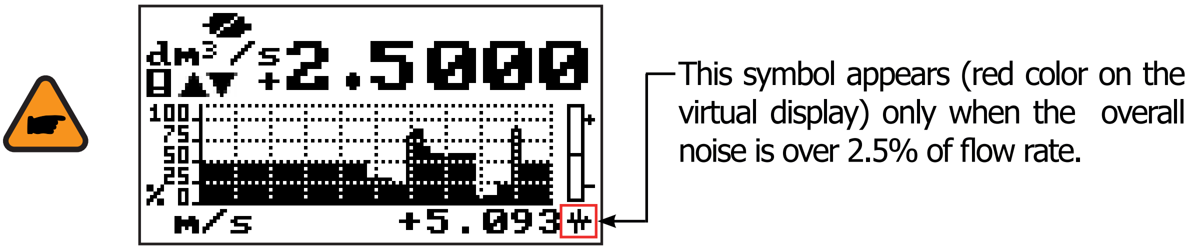
SHORT PRESSING (< 1 SECOND):
Enter /leave the selected function
Enables the main menu for the instrument configuration Cancels the selected function under progress
LONG PRESSING (> 1 SECOND):
Leaves the current menu
Enables the totalizer reset request (when enabled) Confirms the selected function.

ACCESS VIA MCP INTERFACE (VIRTUAL DISPLAY)

MCP is a Windows® software that allows to set all the converter functions and personalize the menu. The MCP program is required for the blind version of the converter. To use MCP interface consult the relevant user manual.



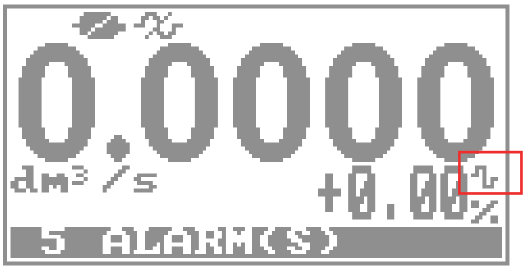
FLOW RATE VISUALIZATION



The MV 210 can show a 5 digits display for flow rate units; this mean the maximum flow rate value that can be represented on the display is 99999 (no matter the positioning of the decimal point). The minimum is 0.0025. The representable measure unit depends on sensor flow rate and diameter; the permitted units are those, that permits the instrument full scale value not exceeding 99999.
Example for DN 300, Full scale value: 3m/s:

- **PERMITTED measure unit** (example): l/s (216.00); m3/h (777.60); m3/s (0.2160)
- **NOT PERMITTED measure unit** (example): l/h (777600)

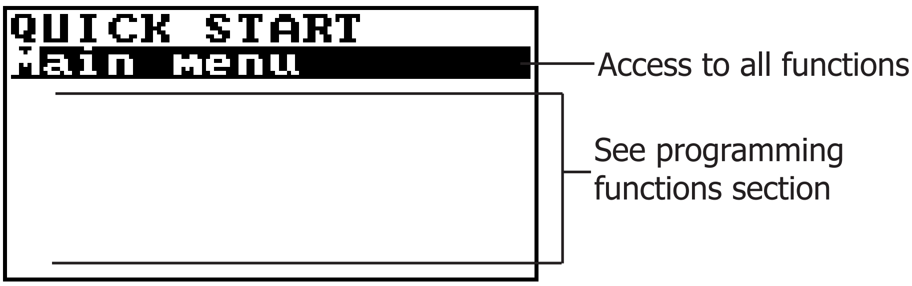
FLOW RATE ALERT



This FLAG becomes active when there is a flow variation (flow rate not stable).

QUICK START MENU

The QUICK START MENU allows to user immediate access to some of the most commonly used functions; through MCP software it possible customize this menu to make it suitable for the specific application.



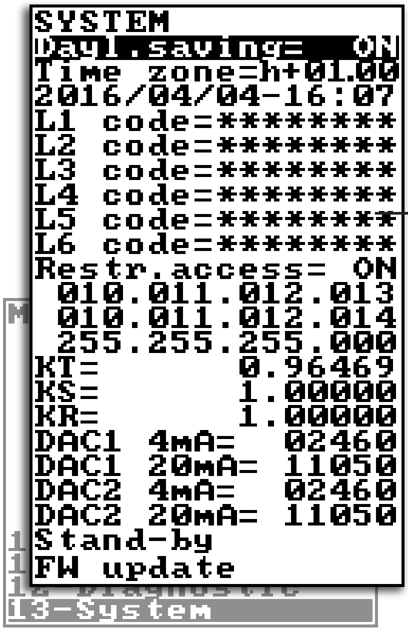
The user has immediate access to the Quick Start menu when the converter is powered up by pressing the Enter key. If access to the quick start menu does not occur, then it could be disabled using the function "9.11" page 29 .

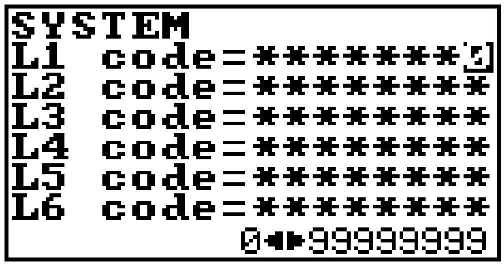
CONVERTER ACCESS CODE

The access for programming the instrument is regulated by six access levels logically grouped.
Every level is protected by a different code.

- ❑ *Access Level 1-2-3-4 Freely programmable by user*

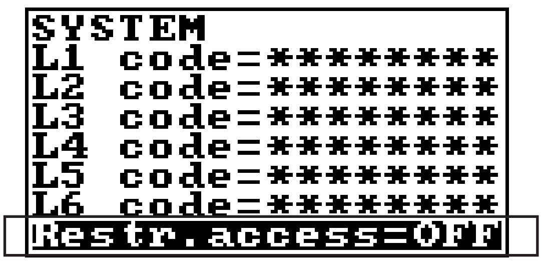
Access Code Set : Menu 13 System





The CODE is Settable by keyboard or MCP interface.
Depending on the level of access different display functions will be visible. (See section "FUNCTIONS DESCRIPTION" page 32)
These access levels interact with the "Restricted access"

Restricted Access Set : Menu 13 System



Settable Values
ON
OFF

Restrict = ON: Access permitted only to functions provided for a specific level;
Example: If the operator has a code of access level 3, after having set it, he can change only the functions with level 3 access.

Restict = OFF: It enables to change functions for the selected level and ALL the functions with lower access level.
Example: If the operator has the code of level 3, after having set it, he can change all the functions at level 3 and those at lower levels.

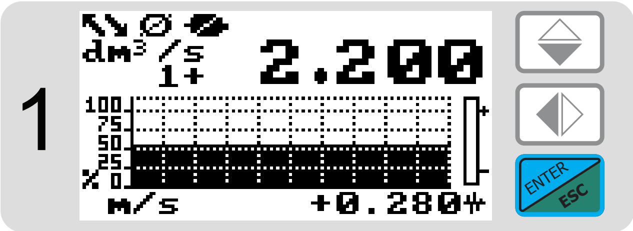
* WARNING: take careful note of the customized code, since there is no way for the user to retrieve or reset it if lost.

Factory preset access codes:

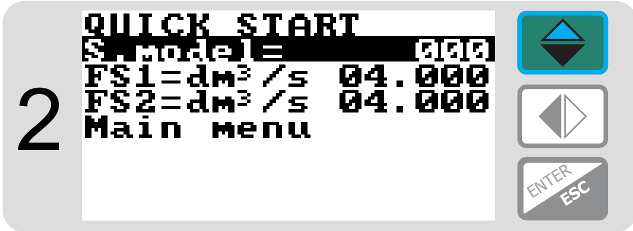
- ❑ **L1: 10000000**
- ❑ **L2: 20000000**
- ❑ **L3: 30000000**
- ❑ **L4: 40000000**

The following example shows how to change the Full scale by Quick Start menu; the second illustrates how to change the function by the Main menu.

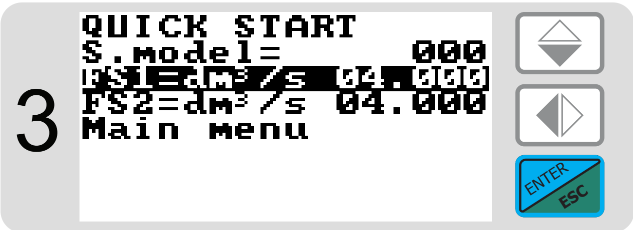
EXAMPLE: modifying the full scale value from 4dm³/s to 5dm³/s, from the "Quick start menu"



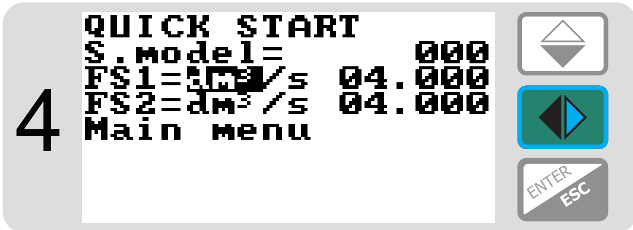
Press the ENTER button to access the Quick Start menu



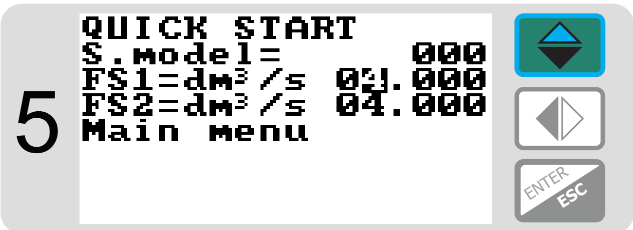
Select this function in the list to be edited



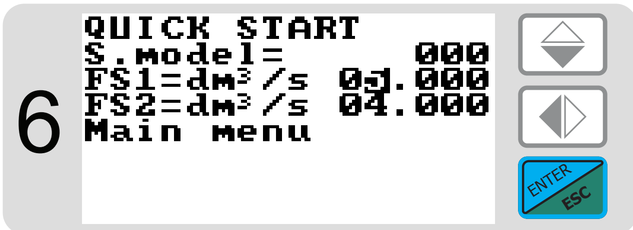
Press the ENTER button to select the function.



Select the value to be changed

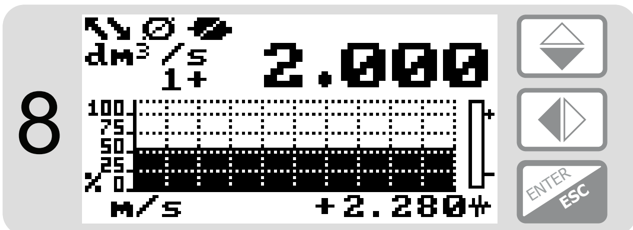


Change the value



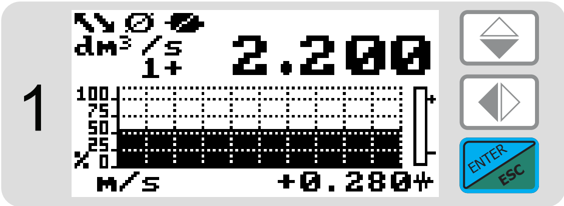
Confirm the new value

Long Push

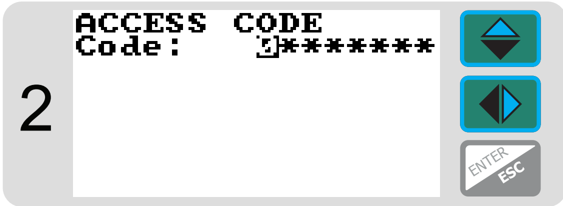


Main Page

EXAMPLE: modifying the full scale value from 4dm³/s to 5dm³/s, from the "Main Menu" (quick start menu enabled)



Press the ENTER button to access the Quick Start menu



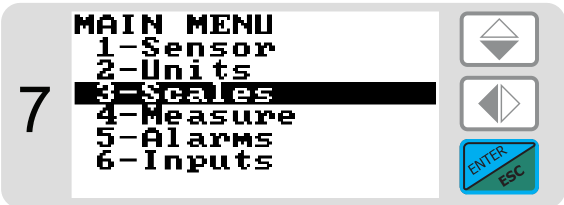
Press arrow keys to select the cell in which to insert the number of the access code.

Press ENTER button to confirm value.

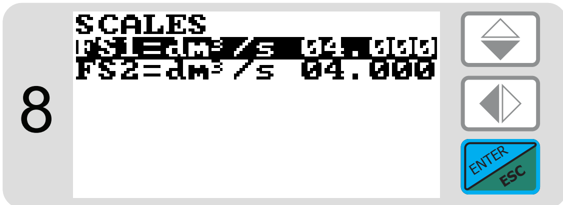
Select "Main Menu"

Press the ENTER button to access the Main Menu

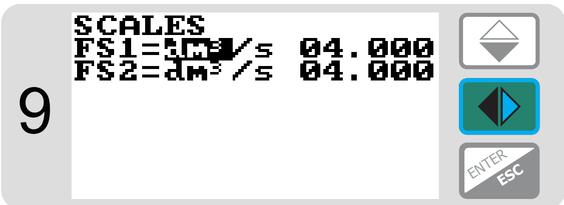
Select function



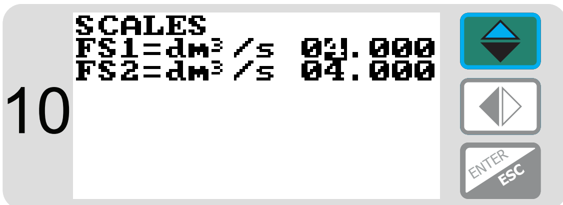
Press the ENTER button to access the"Scale Menu"



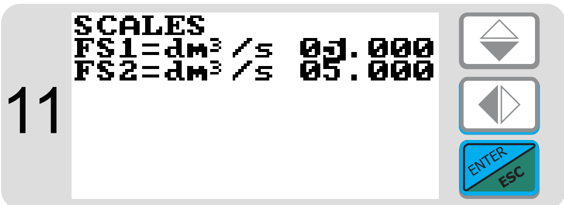
Press the ENTER button to access the "Fs1"



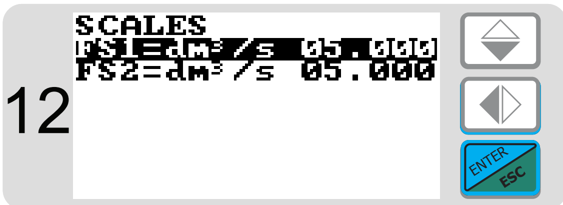
Select the value to be changed



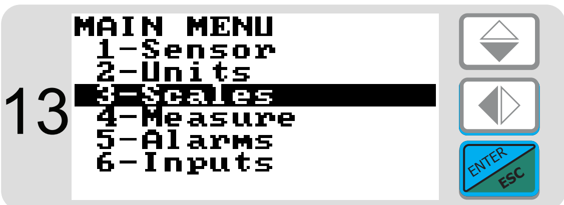
Change the value



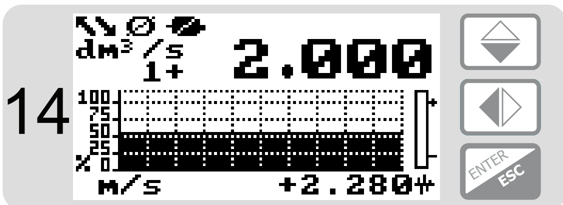
Press the ENTER button Confirm the new value



Press Esc



Press Esc



Main page

FUNCTIONS MENU

The main menu is selected from the Quick start menu by pressing the ENTER/ESC key and entering the access code; enter the access code if required. Note: Some functions are displayed only with other functions active, or with optional modules.

SENSOR

MAIN MENU
1-Sensor

SENSOR

Display	Value	No.	Description
S. model =	0	1.1	Sensors model: Enter the first two characters of the serial number of the sensor
Lining =	UNSPEC.	1.2	Flow sensor lining material type
S. type =	FULL BORE	1.3	Type of sensor: fullbore or insertion
u.type=	METRIC	1.4	Type of measure units for sensor parameter: metric or imperial
Diam mm	00025.0	1.5	Insert ND of sensor (0-2500)
KA =	+00.9637	1.6	Calibration data of sensor visualized on sensor's label
KA- =	-04.4904	1.7	Calibration data of sensor for negative flow
KZ =	+0000000	1.8	Sensor coefficient KZ (zero point)
KD=	+0000000	1.9	Sensor coefficient KD
Ins. position=	0	1.10	Insertion position
KP DinaMmic=	OFF	1.11	KP dynamic, coefficient for insertion
Ki=	01.8727	1.12	Sensor coefficient Ki
Kp=	01.0000	1.13	Sensor coefficient Kp
KC=	1.00000	1.14	Sensor coefficient KC
C.curr =	025.0	1.15	CW volume total. decimal point position
C.Reg.PB=	004	1.16	Current regulator proportional band
C.Reg.DK=	008	1.17	Current regulator derivation constant
S. Freq.= Hz	50	1.18	Measure sampling frequency
Preamplif.	OFF	1.19	Enables the preamplifier
E.P Detect=	ON	1.20	Enables the empty pipe detection feature
R max= kohm	0500	1.21	Empty pipe detection threshold
El. Cleaning=	OFF	1.22	Electrode cleaning
S. cable= m	000	1.23	Sensor connecting cable length
S. err. delay=	010	1.24	Signal error delay (n. sample)
Sens. verify=	OFF	1.25	Automatic sensor verify enable
Zero point cal.		1.26	Pipe hydraulic zero calibration
KL	00.00000000	1.27	Linearization coefficient

UNITS

MAIN MENU
1-Sensor
2-Units
3-Scales

UNITS

Display	Value	No.	Description
Diam.	mm	2.1	Nominal diameter measure unit
S.cable	m	2.2	Cable length on separate version
FR.unit	METRIC	2.3	Flowrate type measure unit: metric or not metric
Pl1 unit	METRIC	2.4	Pulse 1 type measure unit: metric or not metric
Pl2 unit	METRIC	2.5	Pulse 2 type measure unit: metric or not metric
T+ unit	METRIC	2.6	Total direct totalizer measure unit type: metric or not metric
T+ unit	g	2.7	Total direct totalizer measure unit
T+ D.P	3	2.8	Total direct totalizer decimal point position
P+ unit	METRIC	2.9	Partial direct totalizer measure unit type: metric or not metric
P+ unit	g	2.10	Partial direct totalizer measure unit
P+ D.P	3	2.11	Partial direct totalizer decimal point position
T- unit	METRIC	2.12	Total reverse totalizer measure unit type: metric or not metric
T- unit	g	2.13	Total reverse totalizer measure unit
T- D.P	3	2.14	Total reverse totalizer decimal point position
P- unit	METRIC	2.15	Partial reverse totalizer measure unit type: metric or not metric
P- unit	g	2.16	Partial reverse totalizer measure unit
P- D.P	3	2.17	Partial reverse totalizer decimal point position
Temp.unit	°C	2.18	Temperature measure unit
Mass units	ON	2.19	Enable/disable the selection of mass units on full scale set
Sg=kg/dm3	1.0000	2.20	Specific gravity coefficient